Module 2: Perceptron and Artificial Neural Networks (ANN)

1. What Is an Artificial Neuron?

An artificial neuron is a mathematical unit that receives input values, multiplies them by weights, adds a bias, and applies an activation function to calculate an output.

It is loosely inspired by biological neurons, but it does not work exactly like a human brain cell.

2. Components of an Artificial Neuron

An artificial neuron has four important components.

2.1 Inputs

Inputs are the values provided to the neuron.

For example:

- $x_1 = 2$
- $x_2 = 3$
- $x_3 = 4$

For an image, inputs might be pixel values. For a tabular dataset, inputs might be age, salary, or other numerical features.

2.2 Weights

Weights determine how strongly each input contributes to the neuron's calculation.

Each input connection has an associated weight.

For example:

- $w_1 = 0.5$
- $w_2 = 0.8$
- $w_3 = -0.2$

During training, weights are adjusted to help the model make better predictions.

2.3 Bias

Bias is an additional trainable parameter added to the weighted sum.

It helps shift the neuron's response, allowing the model to learn more flexible relationships.

2.4 Activation Function

The activation function transforms the neuron's weighted sum into its output.

Depending on the chosen function, the output could be unrestricted, between 0 and 1, or constrained to another range.

3. Mathematical Representation of a Neuron

The neuron first calculates a weighted sum:

$$\mathbf{z = w_1x_1 + w_2x_2 + ... + w_nx_n + b}$$

Then it applies an activation function:

$$\mathbf{\hat{y} = f(z)}$$

Where:

- x = input value
- w = weight
- b = bias
- z = weighted sum before activation
- f = activation function
- $\hat{y}$ = predicted output

Example Calculation

Suppose:

- $x_1 = 2$
- $x_2 = 3$
- $w_1 = 0.5$
- $w_2 = 0.8$
- $b = 0.1$

Step 1: Calculate the weighted sum.

$z = (0.5 \times 2) + (0.8 \times 3) + 0.1$

$z = 1 + 2.4 + 0.1$

$z = 3.5$

Step 2: Apply an activation function.

If we use ReLU:

$ReLU(z) = \max(0, z)$

$ReLU(3.5) = 3.5$

Therefore, the neuron outputs **3.5**.

4. What Is a Perceptron?

A perceptron is one of the simplest artificial neural models. A classic perceptron is used for binary classification.

It calculates a weighted sum of inputs and applies a threshold function to produce a class prediction.

For example:

Input Features

|

v

Apply Weights

|

v

Add Bias

|

v

Weighted Sum

|

v

Threshold Function

|

v

Class 0 or Class 1

A classic perceptron can learn a linear decision boundary.

However, a single perceptron cannot solve every classification problem. For example, it cannot represent the XOR function correctly.

This limitation motivates the use of networks with multiple neurons and hidden layers.

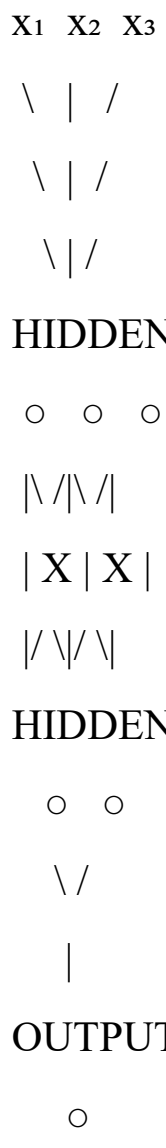
5. What Is an Artificial Neural Network?

An Artificial Neural Network (ANN) consists of interconnected artificial neurons arranged in layers.

A typical feed-forward ANN contains:

1. Input layer
2. One or more hidden layers
3. Output layer

INPUT LAYER



The diagram is illustrative. Real networks can have many neurons and different connection patterns.

6. Understanding the Layers

6.1 Input Layer

The input layer receives the features supplied to the network.

For example, if a dataset has five input features, the model may receive a vector containing five values.

6.2 Hidden Layers

Hidden layers process the information received from the previous layer.

Each neuron generally calculates a weighted sum, adds a bias, and applies an activation function.

Multiple hidden layers allow the network to learn increasingly complex representations.

6.3 Output Layer

The output layer produces the model's final output.

Its structure depends on the task.

Examples:

- Binary classification: one sigmoid output is a common choice.
- Multi-class classification: one output score per class is common.
- Regression: one or more numerical outputs may be used.

7. What Does It Mean When We Say Neurons Are Connected?

In a fully connected or dense network, every neuron in one layer connects to every neuron in the next layer.

Each connection has its own weight.

For example, if a layer has three neurons and the next layer has two neurons, there are:

$3 \times 2 = 6$ connection weights.

The neurons in the next layer also have biases.

These weights and biases are trainable parameters learned during training.

8. Why Do We Need Hidden Layers?

A single perceptron can only learn certain simple relationships.

Hidden layers allow a neural network to combine inputs and learn more complex, non-linear patterns when used with non-linear activation functions.

However, adding hidden layers does not automatically make a model accurate. The model still needs suitable data, training, and evaluation.

9. Perceptron vs. ANN

Perceptron	ANN
Basic artificial neuron or simple perceptron model.	Network of interconnected neurons.
Classic single-layer perceptron has limited expressive power.	Multiple layers can learn more complex relationships.
Classic form uses a threshold activation.	Can use ReLU, sigmoid, tanh, and other activations.
Cannot represent XOR with a single classic perceptron.	A suitable network with hidden layers can represent XOR.

10. Important Clarification

Adding one hidden layer to a simple network gives it a hidden layer; it does not mean every possible ANN must have a hidden layer.

Also, the term "deep learning" commonly refers to learning with neural networks that have multiple layers, although there is no single universal depth threshold used in every context.

11. Quick Revision Questions

1. What are weights and biases?
2. What is the role of an activation function?
3. What is the difference between an input layer and a hidden layer?
4. Why is a single perceptron limited?
5. What is the role of an output layer?
6. What is a trainable parameter?

Module Summary

An artificial neuron calculates a weighted sum of its inputs, adds a bias, and applies an activation function. A perceptron is a simple model, while an ANN connects neurons into input, hidden, and output layers to learn more complex patterns.